Simulation-based cutoffs for conditional item fit in Rasch models: iterations, multiplicity correction, and decision stability

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2026-11-03

Simulation-based cutoff values for conditional item fit can be estimated by parametric bootstrapping and reported as an interval of a chosen width. Johansson (2025) recommended this over rule-of-thumb critical values but gave iteration guidance without accounting for the error rate that the interval width controls. This paper re-examines iterations and width against the familywise error rate (FWER), and compares FWER and false discovery rate control, in seven simulation studies with dichotomous and polytomous items. The interval width is shown to set the FWER of the interval rule jointly with the number of items. Bootstrap p-values with a Westfall-Young step-down target that error rate directly and detect more misfit than a width-matched interval. When all items fit, their FWER was at or slightly below the nominal level from 400 iterations, and the small excess at fewer iterations came from how the statistics are studentized. Holm performed similarly once iterations cleared its floor, which grows with the number of items, while Benjamini-Hochberg flagged more fitting items. When some items misfit, every procedure exceeded its nominal rate at large samples, because misfit contaminates the total score that conditional fit is computed on. Conditional outfit was less powerful than infit throughout. Recommendations include flagging items on the Westfall-Young corrected p-value, using at least 400 iterations and 1000 to 2000 for a final analysis, reporting the interval as a descriptive band at width .95, and removing items one at a time. The methods are available in the R package `easyRasch2` and the jamovi module `easyRasch2jmv`.

Keywords: Rasch, Psychometrics, Item fit, Cutoffs, Critical values, Multiple comparisons, Reproducibility

1 Introduction

Psychometric analysis often relies on rule-of-thumb critical values for interpreting fit metrics, but no fixed value or range can generalize across analyses, because the distribution of a fit statistic under a correctly fitting model depends on features of the data and the model, such as the sample size and the number of items (McNeish & Wolf, 2024; Müller, 2020). Simulation-based critical values for Rasch infit and outfit, obtained by parametric bootstrap, were proposed by Wolfe (2008, 2013) and examined across sample sizes and test lengths by Seol (2016). Johansson (2025) applied this approach to conditional item infit (Müller, 2020) and evaluated it through a series of simulation studies using the `easyRasch` R package (Johansson, 2024). Infit with simulation-based cutoffs outperformed the item-restscore method (Kreiner, 2011) with samples below 500 respondents, and

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the paper provided practical guidance on sample sizes, iterative item removal, and the number of bootstrap iterations. The present paper builds on that work and corrects parts of it.

The iteration guidance in Johansson (2025) recommended 100 bootstrap iterations for samples up to 250 respondents and 200 to 400 iterations for larger samples. That guidance was based on detection rates, since fewer iterations appeared to detect more misfit items with smaller samples. This paper shows that the apparent advantage was an artifact. With few iterations the cutoff interval is estimated from too little information, becomes narrower than its nominal width, and flags fitting items at an inflated rate. Detection was in effect bought with false positives. The false positive rates reported in the earlier paper were calculated per item, while the probability of at least one false flag across all items in a scale, the familywise error rate (FWER), was not examined.

Three contributions follow. First, the interval width and the number of iterations must be chosen jointly. The width sets the FWER of the interval decision rule, and the number of iterations determines whether the nominal width is attained at all. Second, the same bootstrap supports p-values with multiplicity corrections. The Westfall-Young (WY) step-down (Westfall & Young, 1993) targets the FWER directly and controls it when all items fit, detects more misfit than the interval rule at matched error rates, and needs far fewer iterations than the interval rule or the Holm procedure. The Benjamini-Hochberg (BH) procedure (Benjamini & Hochberg, 1995), which controls the false discovery rate instead, is included for comparison and comes with an iteration floor that grows with the number of comparisons. Third, the paper introduces decision stability as an outcome in its own right: whether re-running the same analysis on the same data with a different random seed produces the same set of flagged items.

Results are presented using two simple performance metrics, detection rate and false positive rate, with decision stability added as a third. Readers interested in the statistical foundations of the multiple testing methods are referred to Westfall and Young (1993) and to the more recent paper by Ferreira (2024). All analyses use conditional item fit as implemented in the `iarm` R package (Mueller & Santiago, 2022), accessed through `easyRasch2` (Johansson, 2026a). The bootstrap is parametric throughout. Datasets are simulated from the Rasch model fitted to the observed data, and the model is refitted to every simulated dataset, so the item parameters are re-estimated in each bootstrap sample. The reasons to avoid unconditional item fit statistics, which is implemented in most other software, are detailed in Müller (2020).

The paper proceeds as follows. The next two sections describe the cutoff interval and the decision rules based on bootstrap p-values. The Method section describes the simulation design. Three results sections then cover interval convergence and error rates, the comparison between interval-based and p-value-based decisions, and decision stability. The Discussion summarizes the findings as practical recommendations.

2 The interval and its width

The procedure under study uses a parametric bootstrap to establish critical values for conditional item infit (Müller, 2020). Item thresholds are estimated from the observed data by conditional maximum likelihood, and person locations by weighted likelihood (Warm, 1989). Each of B simulated datasets draws n person locations with replacement from the estimated ones and generates responses under the Rasch model, or the partial credit model for polytomous items (Masters, 1982), with the estimated thresholds, so every simulated dataset fits the model by construction. The model is refitted to each simulated dataset by conditional maximum likelihood, so the item parameters are re-estimated in every bootstrap sample as in the analysis of the observed data, and conditional infit

is computed from that refit. This yields an empirical null distribution of B values per item. Each item's null distribution is summarized by a highest density interval of width w, and an observed infit value outside the interval flags the item as misfitting.

The endpoints of the interval are extreme order statistics of the B simulated values, and how well they can be estimated depends on the expected number of simulated values beyond each endpoint, about $(1 - w)B/2$, here called draws per tail. With B = 400, for example, that number is 10 at w = .95 but 0.2 at w = .999 (Table S1 gives it for other combinations). With fewer than one draw per tail, the interval equals the range of the simulated values, its nominal width is not attained, and its realized width keeps growing as B increases. The number of iterations an interval needs therefore depends on w, and not on the sample size that the iteration guidance in Johansson (2025) was conditioned on.

Once the interval has reached its nominal width, a fitting item is flagged with probability $1 - w$. With k items tested jointly, and treating the item statistics as independent, the probability that at least one fitting item is flagged is

$$\mathrm{FWER} = 1 - w^k, \tag{1}$$

the FWER of the interval rule (Šidák, 1967). The width is thus the error rate parameter of the decision rule rather than a display choice. For a 20-item scale, w = .999 gives a FWER of 2.0% and w = .99 gives 18.2%. Item fit statistics are positively dependent through the shared total score, which makes Equation 1 a slightly conservative upper bound, and the simulation studies below confirm it across item counts and item formats.

3 Bootstrap p-values and multiplicity corrections

The same B simulated values support a different decision rule. Let F_i denote the observed infit of item i and F_{ib}^* the simulated values. Each is studentized by the simulation mean and standard deviation, $t_i = (F_i - \bar{F}_i^*)/s_i^*$, which puts items with different locations on a common scale, since the expected infit range varies with item location relative to the sample (Müller, 2020). This follows Ferreira (2024, Section 4), who standardizes each statistic by its null mean and variance so that the maximum across comparisons treats them equally. In his permutation setting those moments are known and shared by the observed and permuted statistics, while here they are estimated from the simulated values. The marginal Monte Carlo p-value is

$$p_i = \frac{1 + \sum_{b=1}^{B} I(\mid t_{ib}^* \mid \geq \mid t_i \mid)}{B + 1}, \tag{2}$$

where $I(\cdot)$ equals 1 when its argument is true and 0 otherwise.

Its smallest attainable value is $1/(B + 1)$, which sets the resolution of the test. When the observed statistic and the simulated ones are exchangeable under the null, as the observed and permuted statistics of a permutation test are, a p-value of this form is valid at any B (Dufour, 2006, Proposition 2.2). Two features of the procedure prevent that here.

First, the null hypothesis is composite, since the simulated datasets are drawn from the model fitted to the observed data rather than from the model that generated them. Dufour (2006, Section 6) calls such a test, with the nuisance parameters replaced by a point estimate, a local Monte Carlo test, valid asymptotically under regularity conditions rather than at a given sample size. The procedure thus belongs to the bootstrap methods that Ferreira (2024, Section 5) describes as controlling error rates approximately, with an accuracy at finite sample sizes that has to be checked

rather than assumed. Second, the observed statistic is studentized by moments it did not contribute to (mean and standard deviation), while each simulated statistic is studentized by moments that include it, so the test is slightly liberal, by an amount that shrinks as $1/B$. Studentizing all $B+1$ values by moments computed over the observed and simulated values together removes this asymmetry. This joint studentization is the Monte Carlo counterpart of Ferreira's shared moments, and the error-rate study evaluates both versions on the same simulated values.

Either way, the marginal p-value controls the error rate of a single pre-specified comparison. Scanning all k items inflates the familywise error toward the levels in Equation 1, which is the argument for correcting it, not for abandoning p-values.

The WY step-down corrects for multiplicity using the joint distribution that the bootstrap already provides (Ferreira, 2024; Westfall & Young, 1993). The most extreme observed statistic is compared against the distribution of the maximum across all items,

$$\tilde{p}_{(1)} = \frac{1 + \sum_{b=1}^{B} I\left(\max_i |\, t^*_{ib} \,| \geq |\, t_{(1)} \,|\right)}{B+1}, \tag{3}$$

after which the maximum is recomputed over the remaining items for the next statistic, with monotonicity enforced. Only $B \geq 1/\alpha$ iterations, that is 20 at $\alpha = .05$, are needed for a rejection to be possible at all. Ferreira (2024, Section 3) derives the step-down as a shortcut to the closure principle under the subset pivotality condition of Westfall and Young (1993): the joint distribution of the statistics of any set of true hypotheses must not depend on whether the remaining hypotheses are true. The bootstrap reference distribution is that of the complete null by construction, but Ferreira's Example 2, tests of pairwise correlations computed from the same variables, shows that the condition fails when statistics share data, and every conditional fit statistic shares the total score it is conditioned on. The step-down can therefore be expected to control the FWER when all items fit, which is weak control, but not necessarily when some items misfit.

The BH procedure (Benjamini & Hochberg, 1995) instead controls the false discovery rate, the expected proportion of flagged items that are in fact fitting. When a single item stands out, its adjusted p-value is bounded below by $m/(B+1)$ for m comparisons, so it cannot be flagged unless

$$B \geq \frac{m}{\alpha}. \tag{4}$$

When s comparisons have p-values at the Monte Carlo floor of Equation 2, the smallest attainable adjusted p-value is $m/(s(B+1))$, so several clearly misfitting items lower the required B. With 20 items and $\alpha = .05$, a single misfitting item cannot be flagged unless $B \geq 400$.

The Holm procedure (Holm, 1979) is the familiar FWER alternative based on the marginal p-values alone. It controls the FWER under general conditions, needs no pivotality condition, and little improvement on it is possible among procedures that use only the marginal p-values (Ferreira, 2024, Section 2). Procedures that use the joint distribution, such as the step-down, can be more powerful when the statistics are dependent, but the gain is small when they are close to independent, as the agreement with Equation 1 reported below suggests item fit statistics are. Holm shares the reachability floor of Equation 4, which for Holm is not lowered when several items stand out. Ferreira (2024) gives a self-contained account of these methods for 5 to 50 hypotheses, the range of item-level analysis.

All of these rules are computed from one bootstrap and inherit its approximate validity, and the simulation studies quantify how close each comes to its nominal error rate.

4 Method

4.1 Overview

Seven simulation studies were conducted, and Table 1 summarizes their designs. Each answers one question, and its design follows from that question.

The first two use the 20 dichotomous items of Johansson (2025), so that the iteration guidance of that paper is re-examined on the item set it was derived from. The first asks how many iterations the cutoff interval needs at a given width. Convergence concerns the null distribution alone, so it uses null data only, run to B = 10 000 as a reference. The second asks what error rates the decision rules have across widths and iteration counts, with misfit at ρ = .15, the strong misfit of the earlier paper. The third compares the rules on polytomous items at a misfit severity anchored on a real dataset, as described below, so that they are compared at a severity analysts observe rather than one chosen to be detectable. It uses PHQ-9 item parameters for a real configuration and a stylised 15-item set for a regular one. The fourth holds datasets fixed and varies only the bootstrap seed, because decision stability concerns one dataset analyzed twice. It spans misfit from strong to anchored, since stability depends on how many items lie near the decision threshold. The fifth replaces multidimensionality with random responding at matched infit, to check whether two findings belong to the misfit mechanism rather than to misfit in general. The sixth applies the recommended rule and number of iterations to conditional outfit, the second statistic the same bootstrap produces. The seventh repeats the error-rate and rule comparisons on a five-item scale, where Equation 1 predicts the lowest error rates and the reachability floor of BH no longer binds.

Within each replication, the five interval widths (w = .90, .95, .99, .995, and .999) and the decision rules are computed from the same bootstrap, so comparisons between widths and between rules are paired, and in the sixth study so is the comparison between infit and outfit. In the two dichotomous studies every smaller B is evaluated on the first B draws of a single bootstrap per replication, so comparisons between values of B are paired as well, and the two studies share their null replications. Replication counts differ because the studies estimate different quantities with different precision, and the Monte Carlo error of each is reported in Table S7.

Table 1. Overview of the simulation studies

Study	Items	ρ	Misfitting items	n	B	Replications
Interval convergence	20 dichotomous	none	0	150 to 1000	100 to 10 000	500
Error rates over (w, B)	20 dichotomous	.15	0, 1, 3	150 to 1000	100 to 10 000	500 / 250
PHQ-9-anchored misfit	15 polytomous, 9 phq9	.75, .85, .90	0 to 3	150 to 1000	100, 400, 2000	120
Decision stability	15 polytomous, 9 phq9	.15, .50, .85	0 or all	250, 1000	100 to 2000	10 datasets x 25 seeds
Mechanism check	9 phq9	random responses	1, 2	250, 1000	100, 400, 2000	120
Infit versus outfit	20 dichotomous, 9 phq9	.60, .85	0, 1	250, 600	1000	250
Short scale	5 polytomous	.75	0, 1	250, 600	100, 400, 2000	1000 / 250

Note. Misfitting item counts are 1 and 3 for the 20-item and 15-item sets and 1 and 2 for the 9-item set, with 0 denoting the null condition used for error rates. The n = 150 condition is included for the 9-item set only. In the sixth study the two correlations belong to the two item sets in the order listed, chosen so that both inject comparable infit, and its single misfitting item is run as two separate conditions, once near the person mean and once off target. The null replications shared by the first two studies number 500 per sample size. The misfit conditions of the second study run to B = 2000 with 250 replications each. The seventh study uses 1000 replications for its null conditions, which carry the error-rate claim, and 250 for its misfit conditions.

4.2 Data generation

Misfit is induced by multidimensionality, as in Johansson (2025), in every study but the mechanism check. Fitting items are generated from a primary latent dimension and misfitting items from a second dimension, with person locations drawn from a bivariate normal distribution where ρ denotes the correlation between the two dimensions. Lower values of ρ produce stronger misfit. The null conditions, with no misfitting items, use a single dimension.

The mechanism check replaces multidimensionality with random responding, in which a fixed proportion of the responses to the target items is replaced by a uniform draw over the response categories. That proportion was set so that a single target item has the same expected infit as in the multidimensional condition, which puts the two mechanisms on equal footing at matched misfit severity.

The dichotomous item set replicates the design of the earlier paper: 20 items with locations drawn uniformly between −2 and 2 logits, person locations with a standard deviation of 1.5, and response data generated with the `eRm` package (Mair & Hatzinger, 2007). Misfitting items are those located nearest 0, −1, and −2 logits relative to the person mean, so that targeting varies. Johansson (2025) drew every sample from one master dataset of 10 000 simulated respondents per condition. Here every replication of every study generates its respondents anew, so replications

are independent and no chance feature of one master dataset is shared by all the replications of a condition. The sixth study reuses this item set and contrasts two single-item conditions, one misfitting the item nearest the person mean at 0.04 logits and one misfitting the most off-target item at 1.82 logits. This mirrors the q5 and q8 contrast in the PHQ-9 set (see below).

The polytomous item sets use four ordered response categories. The stylised set has 15 items with locations evenly spaced between −2 and 2 logits and item thresholds spread one logit around each location, again with person standard deviation 1.5. The second set takes its item parameters from the phq9 dataset included in `easyRasch2`, nine depression screening items with four response categories answered by 600 respondents. The data are a subsample of the National Health and Nutrition Examination Survey (CDC/National Center for Health Statistics, 2024) in which respondents with all-zero responses were randomly thinned, to avoid the zero-inflated response pattern of the full survey data. In the resulting data 1.3% of respondents have a total score of 0 and the median total score is 16 of 27, so the sample leans toward the upper half of the scale. Zero responses are concentrated in two items, q8 (50%) and q9 (47%), the items located furthest above the person mean. The configuration is therefore unusual in having two off-target items rather than in having a floor effect, and the stylised 15-item set provides a regular configuration for comparison. Thresholds were estimated with conditional maximum likelihood and the latent standard deviation of 1.39 was estimated from the same data. This set represents the item properties of a real instrument rather than stylised ones. Its misfitting items were designated to be q5, located near the person mean, and q8, the most off-target item at 1.4 logits above the mean, so that targeting again varies.

4.3 Calibrating misfit severity

Misfit severity is reported and calibrated on the infit scale rather than on ρ, for two reasons. First, the same ρ produces very different infit depending on item format. Table 2 shows that at identical item locations and person distributions, polytomous items yield roughly 2.8 times the infit excess of dichotomous items, because more response categories carry more information per item. The $\rho = .15$ used in the earlier paper was chosen to represent a clearly misfitting dichotomous item. Carried unchanged into polytomous items it would produce infit values beyond what real data show. Second, infit is the scale on which analysts observe misfit, so calibrating on it connects the simulation conditions to practice.

The same non-portability holds across item counts. A separate scan on the five-item set used in the short-scale study gives a mean injected infit of 1.20 at $\rho = .85$, against 1.36 for the same ρ on the twenty-item polytomous set built to the same recipe, with the nine phq9 items falling between at 1.32. Reaching the anchored range at five items requires $\rho = .75$, which gives 1.32. A short scale needs stronger misfit to show the same infit, because the misfitting item is a larger share of the total score that the statistic is conditioned on and therefore attenuates its own value more. Anchoring on infit rather than on ρ keeps the conditions comparable across item counts as well as formats. We remind readers that infit is strongly dependent on sample size (Müller, 2020) and traditional rule-of-thumb thresholds should be avoided when interpreting infit values.

Table 2. Conditional infit by item format and inter-dimensional correlation

Correlation	Dichotomous	Polytomous	Excess ratio
0.15	1.61	2.71	2.80
0.50	1.38	2.09	2.91
0.75	1.21	1.57	2.72
0.85	1.13	1.36	2.82
0.90	1.09	1.24	2.54

Note. Mean conditional infit of a single misfitting item under identical conditions (20 items, identical locations, n = 500, 60 replications). The excess ratio is the polytomous infit excess over 1.0 divided by the dichotomous excess.

The misfit level used for the rule comparisons, referred to below as the PHQ-9-anchored or anchored level, is set against the phq9 dataset with n = 600. Analyzing the observed responses with simulation-based cutoffs (B = 1000) and WY corrected p-values flags three underfitting items, q3 at 1.23, q8 at 1.26, and q9 at 1.31, along with two overfitting items at 0.78 and 0.83. A calibration scan on the phq9 item parameters shows that ρ = .85 and .90 together reproduce this range, with mean infit values of 1.32 and 1.24 for a single misfitting item, close to the strongest and the mildest flagged real items. The level .75 gives 1.36, slightly beyond the worst real item. The stability study additionally includes ρ = .15 and .50 to span the range from the strong misfit used in the earlier paper down to the anchored level.

Matching mean conditional infit places a misfitting item at the same observed severity on the statistic that analysts see, which is why it was chosen. It does not equate the alternative distributions, the contamination of the conditioning score or the difficulty of detection, and the mechanism check reported below shows that conditions with the same mean infit can produce very different false positive rates. The anchored level is therefore one empirically observed severity, under multidimensionality, and not a representative sample of the misfit that analysts encounter.

4.4 Decision rules and performance metrics

Four decision rules are compared, each computed from the same bootstrap as the others wherever more than one is evaluated. The interval rule flags an item when its observed infit falls outside the highest density interval, evaluated at each of the five widths. The three p-value rules flag an item when its marginal, WY corrected, or BH corrected p-value falls below α = .05, as described in the previous section. Holm, computed from the same marginal p-values, is reported alongside them as a second FWER procedure.

Three performance metrics are reported. The detection rate is the percentage of misfitting items that are flagged. The false positive rate is the percentage of fitting items that are flagged. The familywise false positive rate is the percentage of replications in which at least one fitting item is flagged, which under the null conditions estimates the familywise error rate that Equation 1 describes. The stability metrics are defined in their own section below.

4.5 Software and execution

All simulations were run in R 4.6.1 (R Core Team, 2026) using `easyRasch2` (Johansson, 2026a), which estimates item parameters with conditional maximum likelihood. Results were produced under `easyRasch2` versions 1.1.0 and 1.1.1, and the two dichotomous studies under version 1.4.0. The versions are identical in everything the simulations use, apart from the `statistic` argument

added in 1.1.1 that the infit-versus-outfit study requires. In the polytomous studies every combination of replication and iteration count uses an independent bootstrap. In the two dichotomous studies the values of B are nested within one bootstrap per replication, as described above. The code, the cached results, and the computational setup are described in the online supplement.

5 Interval convergence and error rates

This section reports the two studies that use dichotomous items, together with the null conditions of the polytomous studies. The first question is how many iterations the cutoff interval needs before it attains its nominal width. The second is what the error rate of the interval rule then is, and whether Equation 1 describes it.

5.1 How many iterations the interval needs

Figure 1 shows the median interval width at each number of iterations, expressed relative to the width obtained from all 10 000 iterations of the same bootstrap, the reference width. Two results follow. First, the required number of iterations depends on the width and not on the sample size. Lines for the same width overlap almost exactly across the four sample sizes, with a largest discrepancy of 0.008 at any combination of width and iterations. Convergence is a property of the simulated null distribution and its tails, and the sample size affects where that distribution sits rather than how many values are needed to describe its extremes.

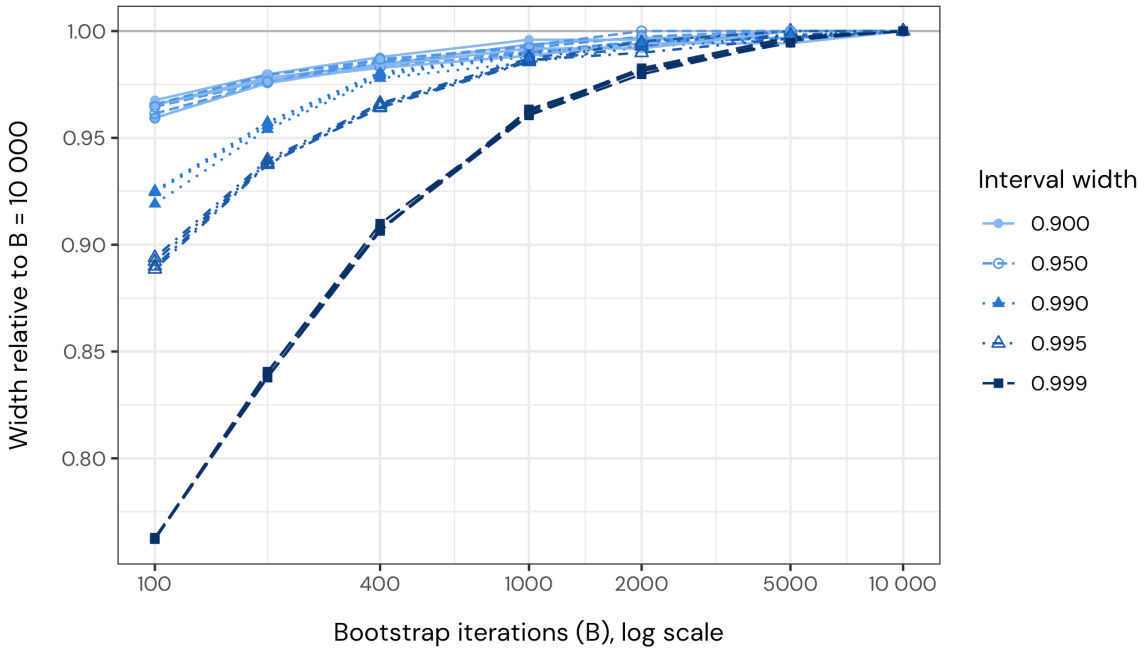


Figure 1. Convergence of the cutoff interval

Note. Median cutoff interval width at each number of iterations, relative to the width at 10 000 iterations of the same bootstrap, for four sample sizes and five interval widths. Each line is one combination of width and sample size, with the width shown by line type and point shape. Lines of the same width overlap, showing that convergence does not depend on sample size. Values below 1 indicate an interval narrower than its nominal width.

Second, the number of iterations required grows sharply as the interval widens. Taking 99% of that reference width as a criterion, 1000 iterations suffice for w = .95, 2000 for w = .99, and 5000 for w = .999. The reference is empirical rather than asymptotic: at w = .999 it has only five draws

per tail itself, so the shortfall reported for that width is if anything understated. At 100 iterations the .999 interval reaches only 0.76 of that reference, that is roughly a quarter too narrow, while the .95 interval is already within 4%. This is the pattern that the expected number of draws per tail, $(1 - w)B/2$, predicts. Widths near 1 place their endpoints far into the tails of the simulated distribution, where few or no simulated values fall, so the interval collapses toward the range of the simulation.

5.2 Error rates of the interval rule

Figure 2 shows the familywise false positive rate under the null conditions, for all three item sets, against the prediction of Equation 1. At iteration counts where the interval has converged, the prediction describes the observed rates well. For the widths .90 to .99 at 2000 iterations, the ratio of observed to predicted rates ranges from 0.82 to 0.99 across all three item sets, with most values slightly below 1. The mild conservatism is expected, since item fit statistics are positively dependent through the shared total score and Equation 1 assumes independence. The interval width therefore sets the familywise error rate of the decision rule, and it does so consistently across 9, 15, and 20 items and across dichotomous and polytomous formats.

The relation also holds below that range, where it is most exposed. On a five-item polytomous scale Equation 1 predicts markedly lower rates of 41.0, 22.6, 4.9, 2.5 and 0.5% for the five widths, so a bound that was loose because of dependence among item statistics would show it here. From 1000 null replications per condition at B = 2000, the observed rates were 37.2, 20.3, 4.6, 3.0 and 0.9% at n = 600 and 39.8, 22.8, 5.8, 3.3 and 0.7% at n = 250. No width departs from its prediction by more than 2.5 Monte Carlo standard errors at either sample size, and at n = 600 the two widest intervals fall below their predictions, the same mild conservatism seen at 9, 15, and 20 items. Equation 1 therefore describes five-item scales as well as longer ones.

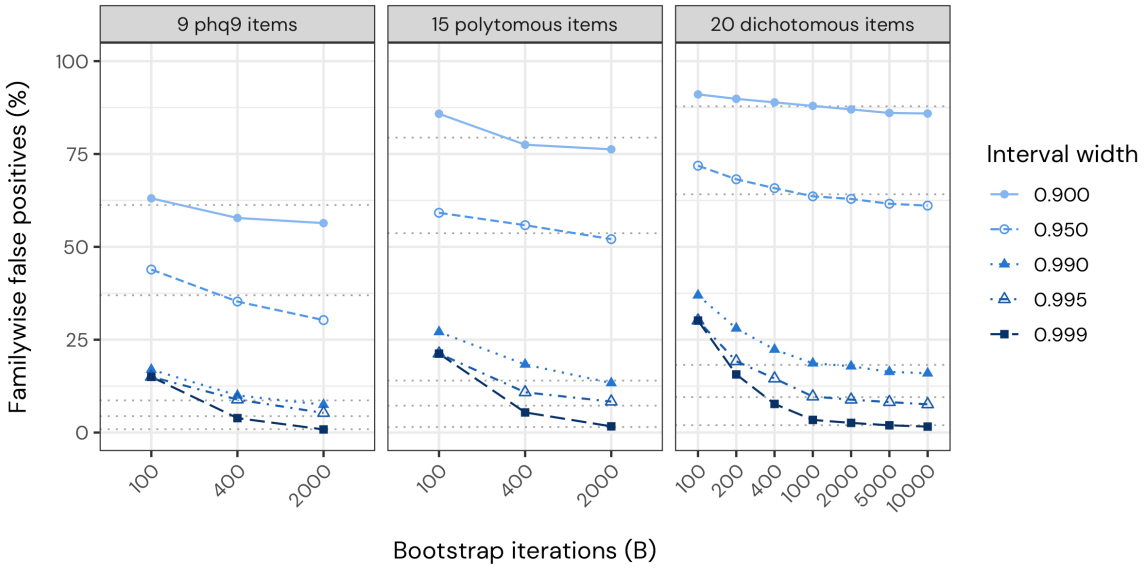


Figure 2. Familywise false positive rate against the prediction of Equation 1

Note. Rates under the null conditions, by interval width and number of iterations, for the three item sets. Dotted grey horizontal lines are the predictions of Equation 1, one for each width. Observed rates approach the prediction as the interval converges, and exceed it when too few iterations leave the interval narrower than its nominal width.

Where the interval has not converged, the observed rates lie above the prediction, and by a wide margin at the narrowest widths. For the 20 dichotomous items at w = .999, whose nominal familywise error rate is 2.0%, the observed rate was 30.1% at 100 iterations and 7.7% at 400. Even at 2000 iterations it remained 2.6%, which is consistent with the interval still being 1.9% narrower than the reference at that point. The two studies thus tell one story from two directions. A small shortfall in interval width translates into a large excess in familywise error, because the error rate is governed by the extreme tails that the shortfall affects.

5.3 The detection and false positive trade-off across iterations

The recommendation of 100 to 400 iterations in Johansson (2025) rested on the observation that detection rates were highest with the fewest iterations. The present results show the same trade-off and explain its source. With three misfitting items among 20 dichotomous items at n = 150, detection across 100, 200, 400, 1000, and 2000 iterations was 84, 79, 72, 67, 67%, a decline of the same magnitude as the one reported in the earlier paper. The false positive rate on the fitting items in the same conditions was 4.2, 2.7, 1.4, 0.5, 0.5%, and the familywise false positive rate under the null fell from 30.1% to 2.6% over the same range. The decline in detection appears only where detection is not already at ceiling. With a single misfitting item in the same item set, detection stayed at or above 99.6% at every number of iterations, and only the false positive rate responded.

Detection and false positives therefore decline together, and for the same reason. Fewer iterations produce a narrower interval, a narrower interval flags more items, and among the flagged items are both misfitting and fitting ones. The apparent advantage of few iterations was not greater sensitivity but a lower and undocumented threshold. Since the earlier paper evaluated false positives per item rather than across the scale, the cost was easy to overlook. A per-item rate of 4% at 100 iterations is unremarkable in isolation, while the corresponding probability of at least one false flag somewhere in a 20-item scale was 30.1%.

6 Interval-based versus p-value-based decisions

6.1 The four rules compared

The p-value rules are first examined under the null conditions of the dichotomous study, where any flag is a false positive. Table 3 shows the familywise error rate by rule and number of iterations, pooled over the four sample sizes. Three patterns stand out. Scanning all 20 items with uncorrected marginal p-values produces a familywise error rate around 60% at every B, which is the expected consequence of testing 20 hypotheses at α = .05 and the reason a correction is needed. The WY step-down is mildly liberal at B = 100, at 5.6%, falls to 4.2% at B = 200, and from B = 400 it is stable and slightly below its nominal 5%, between 3.4 and 3.8% up to B = 10 000. The near-zero Holm and BH rates at B = 100 and 200 are explained by the reachability floor of Equation 4: with 20 comparisons a single standout cannot be flagged below B = 400. BH can flag below that when several items sit at the Monte Carlo floor at once, which is rare under the null, and Holm flagged nothing. From B = 400 Holm was between 2.6 and 3.7% and BH between 2.7 and 3.7%.

Table 3. Familywise error rate of the p-value decision rules

B	Uncorrected	Holm	WY	WY, joint	BH
100	62.2	0.0	5.6	3.5	0.0
200	60.8	0.0	4.2	3.1	0.4
400	60.4	3.7	3.8	3.4	3.7
1000	60.6	2.6	3.4	3.2	2.7
2000	61.3	3.3	3.5	3.4	3.4
5000	60.5	3.1	3.5	3.5	3.2
10000	60.5	3.2	3.4	3.4	3.4

Note. Percentages under the null conditions of the dichotomous study (20 items, no misfitting items), by number of bootstrap iterations. Pooled over the four sample sizes, 2000 datasets per cell. Uncorrected is the marginal p-value of Equation 2. WY is the Westfall-Young step-down, WY, joint the same step-down with the observed and simulated statistics studentized together over all $B + 1$ values, computed from the same bootstrap, and BH the Benjamini-Hochberg procedure. All values of B within a replication come from one bootstrap of 10 000 iterations.

The decline of the WY rate between B = 100 and 400 is the studentization effect described in the section on bootstrap p-values. With joint studentization the rate was between 3.1 and 3.5% at every B, and paired within replications the standard version exceeded the joint one by 2.0 percentage points at B = 100 (SE 0.3) and by at most 0.1 points from B = 1000. The excess at small B is therefore a property of the studentization rather than of the step-down.

What remains is a level slightly below 5% that does not change between B = 400 and 10 000 under either version, and the per-item marginal tests flagged 4.6 to 4.8% of fitting items at every B. An error that does not shrink as B grows is not Monte Carlo error but the finite-sample departure that Ferreira (2024, Section 5) notes for bootstrap methods with estimated parameters, here conservative. It was smaller in the polytomous studies, at 4.2 and 4.7% from B = 400.

The reachability floor also shapes detection. With a single misfitting item at n = 150, BH detection across B = 100 to 2000 was 0, 7, 100, 100, 100%, jumping exactly where B crosses the floor of 400. With three misfitting items the sequence was 17, 76, 83, 82, 84%, with substantial detection already at B = 200, because three p-values at the Monte Carlo floor lower the attainable adjusted p-value to $m/(3(B + 1))$. Holm detected nothing below its floor in either condition (0, 0, 100, 100, 100% and 0, 0, 71, 72, 76%), since several standouts do not lower it. The WY rule was flat across the whole range at 100, 100, 100, 100, 100% and 77, 78, 78, 77, 77% respectively, since it needs only $B \geq 20$ to reject and its power does not depend on B beyond that.

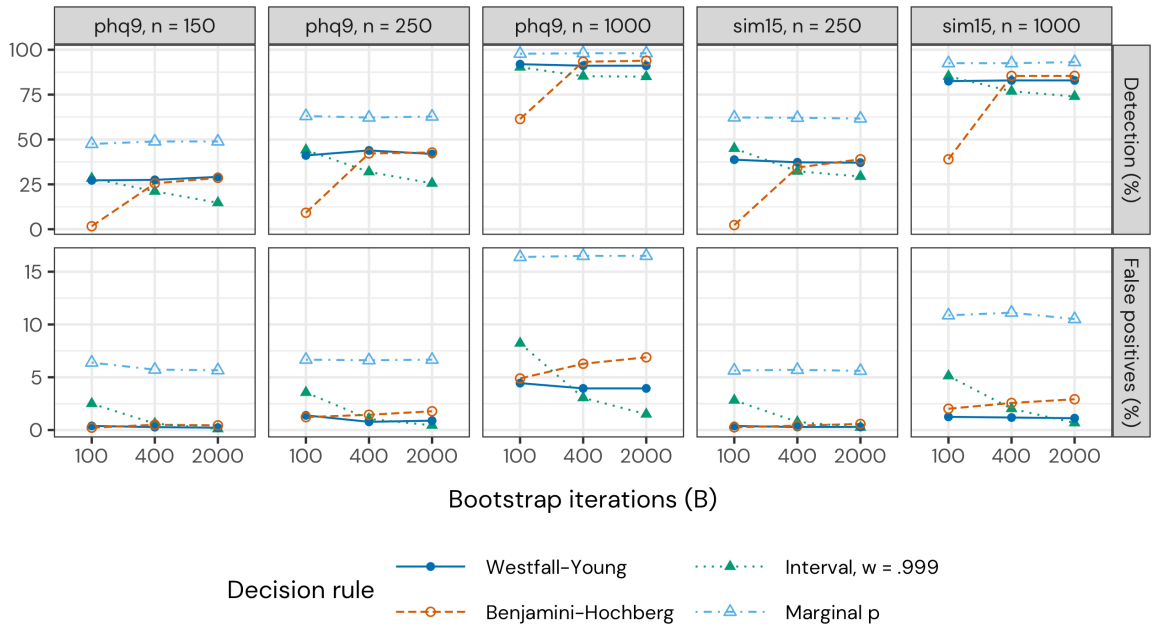


Figure 3. Detection and false positives for the four decision rules

Note. Detection rate (misfitting items) and false positive rate (fitting items) at the anchored misfit level, by number of iterations, item set, and sample size. Rates are averaged over the misfitting-item conditions. Note the different y-axis scales for the two rows.

Figure 3 compares all four rules at the anchored misfit level. Against the interval at $w = .999$ the WY rule detects substantially more misfit, by 8.8 to 18.3 percentage points across the non-saturated nine-item conditions at $B = 2000$, at a false positive rate of 1.2% against 0.6% for the interval, pooled across arms, so it detects more at a higher cost. Part of this raw gain, however, reflects the interval's implicit error rate rather than the method. By Equation 1, $w = .999$ corresponds to a familywise rate of only 0.9% for nine items, so the interval is operating far below the $\alpha = .05$ of the p-value rules.

A fair comparison holds the error rate constant. For nine items, $w = .995$ implies a familywise rate of 4.4% by Equation 1, and its observed null rate at $B = 2000$ was 5.3%, closely matching the p-value rules. Against this matched interval the WY rule still detected more misfit in every non-saturated condition, by 1.7 to 5.0 percentage points, with slightly fewer false positives. The remaining advantage comes from studentization and from using the joint distribution of the item statistics. More important than its size is what the matching exercise reveals: an interval width that matches a chosen α must be solved from Equation 1 for each item count, and for most item counts no convenient width exists. The WY rule targets α directly, whatever the number of items.

Table S4 reports Holm by condition. At $B = 100$ it detected 0.0% of the misfitting items, since the floor of Equation 4 is 180 iterations for nine items and 300 for fifteen. At $B = 2000$ the WY rule detected between −1.7 and 1.7 percentage points more than Holm across conditions, paired within replications, with standard errors of up to 2.0 points, and the pooled false positive rates were 1.2% against 1.0%. Using the joint distribution of the item statistics therefore adds little power here, as Ferreira's (2024) account leads one to expect for statistics that are close to independent. The practical advantage of the step-down over Holm is that it needs no minimum number of iterations.

A short scale tests BH where the floor does not bind at all. At five items the floor is 100 iterations, so the procedure operates freely at every B examined, and the comparison is no longer handicapped by reachability. At $n = 250$ it detected the misfitting item in 91.2% of replications

against 90.8% for the WY rule, with false positives of 6.6% against 4.2%. At n = 600 both detected it in every replication, with false positives of 22.7% against 16.3%. Paired within replications, the detection difference was within one standard error of zero at both sample sizes while the false positive difference exceeded four. The preference for the WY rule is therefore not an artifact of testing BH only where its floor binds.

BH and WY performed similarly with a single misfitting item, while with several misfitting items BH detected somewhat more, as expected since the false discovery rate is a more permissive criterion when true effects are present. The price appears on the other side of the ledger, with pooled false positives of 2.3% against 1.2% at B = 2000, roughly a doubling. The marginal p-value, finally, is close to its nominal level per item but unusable as a screening rule, as its familywise rate of around 60% shows. Once items misfit its false positive rate on the fitting items rises, which the section on error rates under misfit returns to. Taken together, these results support the WY p-value as the default among the decision rules examined, with Holm a close alternative once B clears its floor, the marginal p-value reserved for a single pre-specified item, and BH for longer instruments where a false discovery criterion is acceptable and B clears the reachability floor. The interval retains descriptive value, which the Discussion returns to.

6.2 Conditional outfit under the same rules

Every result so far concerns conditional infit. The same bootstrap produces conditional outfit at no additional cost, and the decision rules are indifferent to which statistic they are given, so it is worth asking whether anything is gained by reading both. Outfit is also widely reported in its unconditional form, for example by RUMM2030, a widely used Rasch analysis software package. The comparison here concerns conditional outfit only, and its results need not carry over to the unconditional statistic, whose behavior under the model differs (Müller, 2020). Johansson (2025) compared conditional infit and outfit for dichotomous items under the interval rule and found outfit worse throughout. The sixth study revisits that comparison under the corrected p-value and extends it to polytomous items. The two item sets were placed at correlations that inject comparable infit, 1.28 for the dichotomous set and 1.27 for the phq9 set, both inside the range the real phq9 items reach. A single misfitting item was placed either near the person mean or off target, the WY rule was applied at B = 1000 and α = .05, and both statistics were read from the same bootstrap within each replication.

Under the null conditions the FWER was 2.4 to 4.8% for outfit against 3.6 to 6.4% for infit, across the four combinations of item set and sample size, with a Monte Carlo standard error of 1.4 points. The marginal rates before the step-down were 3.8 to 4.9% and 4.4 to 5.3%. Neither the statistic nor the correction is miscalibrated, so the studentized maximum needs no modification when the statistic it is given changes.

Detection is a different matter. Infit detected the misfitting item more often than outfit in every one of the eight conditions, and Table 4 gives the margins. The gap is not small, ranging from 4 to 41 percentage points. Outfit is the unweighted mean of squared standardized residuals and is conventionally expected to earn its place on items far from the person mean, and the injected outfit is indeed larger there. Detection nevertheless moves the other way. Averaged over item sets and sample sizes the infit advantage was 16 points for the on-target item and 33 points for the off-target one. The null conditions explain why. Outfit is the more variable statistic under a true model, by a factor of 2.9 in the dichotomous set and 1.4 in the phq9 set, and within the dichotomous set that

factor is 2.2 for the five most central items and 3.6 for the five most extreme. The variance grows faster than the signal exactly where outfit was expected to help.

Because both statistics come from the same bootstrap, the comparison can be made replication by replication rather than only on the margins. Table 4 reports the discordant counts. Replications in which outfit flagged the misfitting item and infit did not never exceeded 14 of 250, against up to 108 in the other direction, and every condition favours infit at $p < .01$ by an exact test on the discordant pairs. In these conditions outfit added little to what infit detected.

This has a direct consequence for practice, because both statistics often appear side by side in the standard output. Flagging an item when either statistic is significant tests $2k$ hypotheses while each correction covers k, and the null FWER of that combined rule was 6.4 to 8.8%, against a nominal 5% and the 2.4 to 6.4% the two statistics reach on their own. The inflation is below the doubling that independent statistics would produce, since the two are computed on the same responses and are positively dependent, but it is real and adds almost no detection. For conditional statistics at these settings, the results favor deciding on infit and treating outfit as descriptive.

Table 4. Detection and paired disagreement, conditional infit versus outfit

Items	Target	n	Det infit	Det outfit	Only infit	Only outfit	p
20 dich	On target	250	83.6	44.0	101	2	< .001
		600	100.0	94.4	14	0	< .001
	Off target	250	59.2	18.4	108	6	< .001
		600	94.4	58.4	92	2	< .001
9 phq9	On target	250	69.2	55.2	41	6	< .001
		600	98.8	94.8	10	0	0.002
	Off target	250	52.4	24.8	83	14	< .001
		600	94.0	66.8	73	5	< .001

Note. Westfall-Young (WY) rule at $B = 1000$ and $\alpha = .05$, 250 replications per condition. Det is the percentage of replications in which the misfitting item was flagged. Only infit and Only outfit count the replications in which one statistic flagged it and the other did not. The p-value is an exact binomial test on those discordant counts. Design and false positive rates are in Table S6.

6.3 Random responding under the same rules

Every result so far induces misfit through multidimensionality, so two of them could in principle be properties of that mechanism rather than of misfit in general. The first is the false positives that appear on fitting items once misfitting items are present. The second is the masking that weakens detection when two items misfit together. The mechanism check separates the two. It holds the item set, the conditions and the decision rules fixed, and replaces multidimensionality with random responding at matched infit.

The false positives are not specific to multidimensionality. With two misfitting items at $n = 1000$ and $B = 2000$, random responding produced false positives on the seven fitting items of 11.3% under the interval rule at $w = .999$ and 22.5% under the WY rule, against 1.7% and 4.8% under multidimensionality. Nearly all false flags pointed in the overfit direction under the interval rule for both mechanisms, at 100% and 97% of flags. Direction is read from the band, so the p-value rules cannot be scored the same way. What puts a fitting item outside its expected range is contamination of the total score that it is conditioned on, rather than the particular way that contamination arises.

The error rate results are therefore not an artifact of the mechanism, and the multidimensional condition is the more forgiving of the two.

Masking is mechanism-specific, and should be read as such. Two independently noisy items do not mask each other, and detection of the two target items at n = 250 was 55.4% under random responding against 31.7% under multidimensionality. Masking requires the misfitting items to share a source. That is the case worth planning for, since a secondary dimension rarely involves a single item, but it is not a property of misfit in general.

6.4 Error rates when some items misfit

The error rates reported so far were measured under the null. With misfitting items present, strong FWER control concerns the probability of flagging at least one fitting item, and BH targets the false discovery rate, the expected proportion of flagged items that in fact fit. Table S5 reports both for every condition with misfitting items, at B = 2000.

At sample sizes of 250 or below the rates were close to nominal. At the anchored misfit level in the polytomous sets, the WY rate over the fitting items was between 0.8 and 8.3%, and the false discovery rate of BH between 1.7 and 7.0%. At n = 1000 every procedure exceeded its nominal level. For the nine phq9 items, the WY rate over fitting items was 25.0% with one misfitting item and 30.0% with two, the Holm rate 21.7% and 26.7%, and the false discovery rate of BH 18.3% and 19.5%. The rates grew with misfit severity, reaching 72.5% for WY at ρ = .75, and were higher under random responding, at 82.5% with two misfitting items at n = 1000.

The source of the excess lies in the per-item tests rather than in the corrections. At n = 1000 the uncorrected marginal p-value flagged 13.6% of the fitting phq9 items with one misfitting item and 19.8% with two, where a valid per-item test would flag about 5%. Each fitting item's statistic is conditioned on a total score that includes the misfitting items, so its distribution changes when they misfit, while the bootstrap reference is generated from a model in which all items fit. This is the failure of subset pivotality described earlier, and Ferreira (2024, p. 2) notes that a p-value can have the wrong distribution because the model is incorrect rather than because its null hypothesis is false. It also explains why Holm, which needs no pivotality condition but relies on valid marginal p-values, failed in the same conditions. The procedures studied here therefore provide weak FWER control, and no correction applied to these per-item tests can provide strong control once misfit contaminates the conditioning score. The contamination grows with sample size and misfit severity, and removing the most misfitting item before judging the rest reduces it.

The false discovery rate of BH stayed close to nominal where the per-item tests were close to valid and exceeded it where they were not, and the WY rule had a false discovery rate at or below it in every condition. The preference for a familywise criterion rests on the use of the analysis. A scale of 5 to 20 items is judged item by item, each flag leads to an item being considered for removal, and a single false flag changes the instrument. The false discovery rate is most useful when many hypotheses are tested and many discoveries are expected (Ferreira, 2024), which is rarely the case in item fit analysis.

7 Decision stability

The previous sections report error rates averaged over many simulated datasets, but an analyst has one dataset, and two analysts who run the same analysis on it with different random seeds can reach different conclusions. To quantify how often, ten datasets per condition were held fixed and each was analyzed with 25 bootstrap seeds. The observed infit values are then fixed, and all variation between seeds is simulation noise.

For each item and dataset, the flag rate p is the proportion of the 25 seeds in which the item was flagged, and an item is counted as unstable when $0 < p < 1$. Twenty-five seeds resolve p in steps of 0.04, so the unstable counts below are lower bounds. Two analyses disagree when their sets of flagged items differ, and the probability of this is measured directly as the proportion of the 300 pairs of seeds that disagree,

$$D = 1 - \frac{\sum_j \binom{c_j}{2}}{\binom{S}{2}}, \tag{5}$$

where c_j is the number of seeds that produced the j-th distinct set of flagged items and $S = 25$ is the number of seeds. D is computed within each dataset, averaged over the ten datasets of a condition, and needs no assumption about how the items depend on each other. For a single item the corresponding probability is $2p(1-p)$, the chance that one of two seeds flags the item and the other does not.

Under the null conditions, flagging with the WY p-value is essentially reproducible. Pooled across both item sets and sample sizes, 1.2% of items had a seed-dependent flag at $B = 100$, 0.2% at $B = 400$, and none at $B = 2000$. At the anchored misfit level ($\rho = .85$), pooled the same way, 3.3% of items were unstable at $B = 400$ and 1.0% at $B = 2000$ (Monte Carlo standard error 0.7 percentage points), against 10.6% and 6.0% for the interval rule at $w = .999$. Pooled at the anchored level, the probability that two seeds using the WY rule disagree about at least one item was 23% at $B = 100$, 10% at $B = 400$, 6% at $B = 1000$, and 4% at $B = 2000$. $B = 400$ is thus where the error rates are trustworthy, but reproducibility keeps improving well past it.

Treating the items as flipping independently, with each $2p(1-p)$ multiplied by $S/(S-1)$ to remove the bias of estimating p from 25 seeds, gives nearly the same pooled values, within 0.2 percentage points. The average hides where independence fails. For the WY rule the independence product overstated the measured disagreement by more than 5 percentage points in 44 of 480 combinations of dataset and number of iterations, by up to 37 points, and never understated it by as much. In those datasets one seed's maximum statistic moves several decisions together, the coupling that the step-down introduces. For the interval rule, whose items are judged separately, the two agreed within 8 points in every dataset.

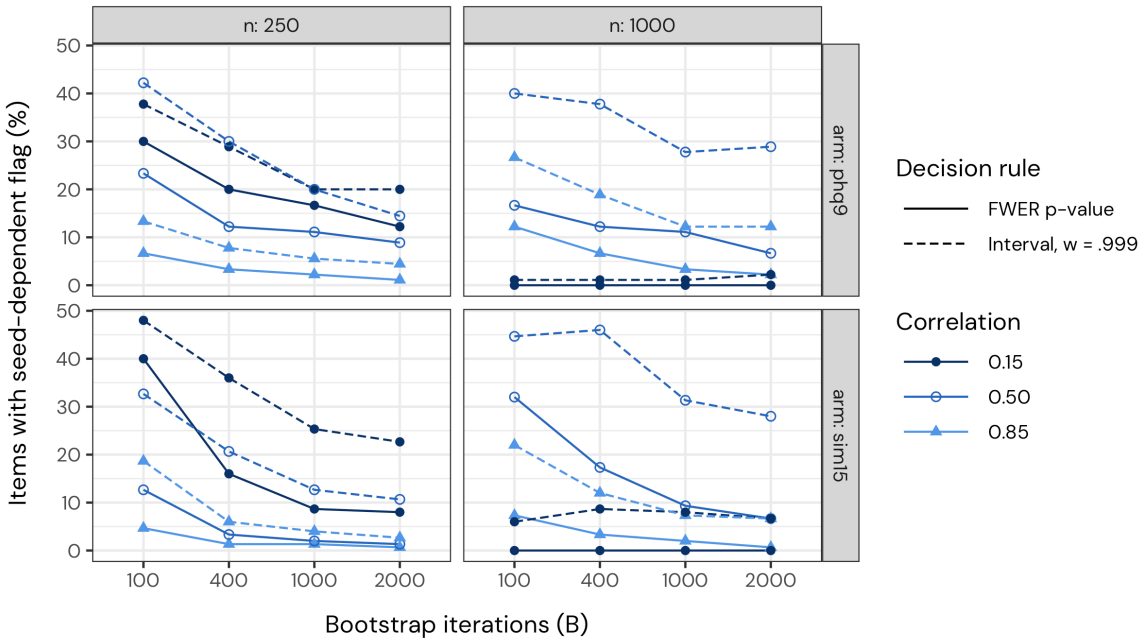


Figure 4. Seed dependence of flagging decisions

Note. Percentage of items whose flagging decision depends on the bootstrap seed (25 seeds per dataset, 10 datasets per condition), by number of iterations, misfit severity, decision rule, item set, and sample size. Lower values of the inter-dimensional correlation mean stronger misfit. Per-condition values carry Monte Carlo standard errors of up to about 3 percentage points and are tabulated in Table S8.

Figure 4 shows that instability is not monotone in misfit severity or sample size. At B = 2000 the strongest misfit (ρ = .15) left on average 1.1 items per dataset with a seed-dependent flag at n = 250 but 0.0 at n = 1000, where items were flagged or cleared decisively. Instability arises where the evidence about an item approaches the flagging threshold without clearly passing it, so it is driven by the number of such boundary items rather than by misfit severity or sample size as such. Across the misfit conditions, the probability that two seeds disagreed at B = 2000 ranged from 0 to 23%. With ten datasets per condition, these results illustrate the dependence rather than giving rates that carry over to other data.

How much instability an analysis shows is thus set by the data. If an item's p-value with unlimited iterations is π, its Monte Carlo estimate from B iterations has a standard error of about $\sqrt{\pi(1-\pi)/B}$, so the iterations needed for a reproducible decision grow with the inverse square of the distance between π and α, and for items close to the threshold no affordable number of iterations removes the instability. Seed sensitivity is therefore a diagnostic. Items whose flags change with the seed are borderline cases that need more data or substantive judgment rather than a luckier seed. The instability of the interval rule at w = .999 declined only slowly with B, consistent with its extreme quantiles remaining poorly estimated throughout the practical range.

8 Discussion

This paper asked how the interval width and the number of bootstrap iterations should be chosen for simulation-based cutoffs, and whether the interval is the best way to turn a bootstrap into a decision about item fit. The answers are connected. The width is the FWER parameter of the interval rule and is described well by Equation 1. The number of iterations required is set by the

width, and not by the sample size that the earlier guidance conditioned on. The same bootstrap also supports a p-value rule that targets a chosen error rate directly, detects more misfit at a matched error rate, and needs far fewer iterations.

Detection at the anchored misfit level was modest. With the nine phq9-derived items and a single misfitting item at ρ = .85, the WY rule detected the misfit in 62.5% of replications at n = 250 and 50% at n = 150, reaching 100% only at n = 1000, and the interval rule detected less. Two misfitting items were harder to detect than one. When q8 misfit alongside q5, detection of q5 at n = 250 fell from 62.5% to 47.5%, because items that share a secondary dimension are evaluated against person estimates contaminated by that dimension. At n = 250 or below, an item fit analysis that flags nothing is therefore weak evidence that nothing is wrong, and other methods of evaluating fit should be used alongside it.

One conclusion of Johansson (2025) is confirmed and extended. That paper found conditional outfit worse than conditional infit for dichotomous items under the interval rule. Under the WY rule at B = 1000, conditional outfit was calibrated but detected less misfit than infit in every condition studied, for polytomous items as well as dichotomous ones. Its weakness is one of power, and it follows from its larger variability under a true model, which grows with distance from the person mean, so the off-target items that outfit is conventionally recommended for are the ones it handles worst.

8.1 Recommendations for practice

Five recommendations follow from the results.

1. **Decide with the corrected p-value.** Among the decision rules examined, the WY step-down at α = .05 performed best, and it is the recommended decision rule for conditional item fit. Its FWER under the null was at or slightly below the nominal level from B = 400 onward. Holm gave nearly the same results once B cleared its floor of k/α iterations, but the step-down needs no such floor. It also detected more misfit than the interval rule at a matched error rate, and it was the more stable of the two under reseeding. Against BH it holds up even where that procedure is unconstrained: on a five-item scale, whose reachability floor of 100 iterations binds at no B examined here, the two detected the misfitting item equally often while BH flagged roughly half again as many fitting ones. The control holds when all items fit, and once items misfit the false flags on fitting items grow with sample size for every rule, which the fifth recommendation addresses.
2. **Use at least 400 iterations, and more when computation allows.** Below 400 the WY rule was mildly liberal under the null, at 5.6% at B = 100, and decisions were noticeably seed-dependent. The probability that two analysts using different seeds disagree about at least one item kept falling past that point, from 10% at B = 400 to 4% at B = 2000, so at least 1000 to 2000 iterations are worth the cost for a final analysis. Neither the sample size nor the number of items sets this requirement. The step-down can reject from $B = 1/\alpha$ = 20 at any scale length, its null rate from B = 400 was between 3.4 and 5.0% across the 5, 9, 15 and 20-item sets with no trend in the number of items, and the floor of 400 is set by calibration and reproducibility. Holm and BH, in contrast, cannot flag even a single standout item below k/α iterations, a floor that grows with the number of items.
3. **Report the interval as a description and set its width to .95.** At w = .95, 1000 iterations bring the interval within 1% of the width it reaches with 10 000, so the band shown to readers covers close to the 95% it claims. At w = .999 the same 1% takes 5000 iterations, far more than is

usually run, so the stated width is not the width the reader is actually seeing. A band showing where a fitting item's infit is expected to fall is worth reporting, but only when its stated width is real.

4. **Report the number of iterations and the random seed.** Both are part of the analysis. Without them a reader cannot tell whether a borderline item would have been flagged in a rerun. The seed will however only be useful if the data are also shared.
5. **Remove items one at a time.** This is the workflow recommended by Johansson (2025), and its sequential error rates were not evaluated here, since every reported rate refers to a single analysis. It is retained because misfitting items contaminate the total score that the other items are judged against, which is the source of the false flags on fitting items when some items misfit, and removing the most misfitting item first reduces that contamination before the remaining items are judged. Repeated testing across removal rounds adds multiplicity that the corrections studied here do not address, as the Limitations discuss.

Versions of `easyRasch2` before 1.2.0 used w = .999 and B = 250 by default, a setting at which the interval equals the range of the simulated values. From version 1.2.0 the package flags items on the WY p-value, shows the interval at w = .95 and runs 400 iterations by default, and the upcoming version 1.4.1 studentizes the observed and simulated statistics jointly. These recommendations are not confined to users of R. The same analyses are available through `easyRasch2jmv` (Johansson, 2026b), a module for the jamovi statistical platform (The jamovi project, 2026) that delegates every computation to `easyRasch2`, so that results are numerically identical given the same seeds and iteration counts. Simulation-based cutoffs can be more demanding to use than a rule of thumb, since they require a decision about the number of iterations and about which decision rule to apply, and the recommendations given here are meant to make those decisions simpler.

9 Limitations

Misfit was induced by two mechanisms only, multidimensionality throughout and random responding in the mechanism check. Other mechanisms, such as local dependence and genuine variation in item discrimination, were not examined.

Each replication is a single pass through the data. No study simulates the iterative workflow in which a flagged item is removed and the remaining items reanalyzed, which is how the procedures are used in practice and the response to their lack of strong control. Its error rates, including the multiplicity added by repeated testing across removal rounds, are the most important open question.

The scope is narrow in further respects. The item sets had 5, 9, 15, and 20 items, so Equation 1 and the iteration requirements are verified in that range and not for longer instruments. The comparison of conditional infit and outfit was made at one decision rule and iteration count, the WY p-value with B = 1000, and does not necessarily extend to the unconditional outfit that some software reports. Of the FWER procedures only the WY step-down and Holm were compared, and resampling-based false discovery rate procedures (Yekutieli & Benjamini, 1999) were not included. Benjamini-Hochberg is guaranteed to control the false discovery rate under positive regression dependency (Benjamini & Yekutieli, 2001), but whether that extends to two-sided tests on a residual-based statistic such as conditional infit is open, so its rates here rest on the simulations alone. Joint studentization was evaluated in the dichotomous null conditions only. The two versions differ only in whether the observed value contributes to the moments, a share of $1/(B + 1)$, and in those conditions the difference in familywise error was 0.5 percentage points at B = 400 and

at most 0.1 from $B = 1000$. A similarly small difference is expected in the polytomous sets and under misfit at the recommended numbers of iterations, but this was not verified. Exact validity at a given sample size, by maximizing the Monte Carlo p-value over the nuisance parameters (Dufour, 2006) or, for dichotomous items, by conditioning on the observed margins (Verhelst, 2008), was not examined. Every study used $\alpha = .05$. The reachability floors and the correspondence between width and familywise error hold at any α, but the reported rates and the 400-iteration floor were established at .05, and a smaller α would likely require more iterations. The anchored misfit levels were calibrated against a single instrument with an unusual configuration, two items far above the person mean, and the stylised 15-item set is the only regular configuration it is compared with. Matching on mean infit, finally, does not make different misfit mechanisms equivalent, as the mechanism check shows.

A practical limit appears at the short end. Five polytomous items with four response categories imply fifteen threshold parameters, and at $n = 250$ 2 of the 3748 bootstraps run in that study produced no usable p-value at all, their simulated statistics having too little variance to studentize by. This did not occur at $n = 600$ or with 9 items or more, but a five-item scale at a small sample is where the procedure starts to strain.

10 Use of large language models

Large language models were used in preparing this work. The author designed the studies, decided which conditions to run and which results to report, verified the code and the reported values, and is responsible for all content including any errors.

A large language model was used in an agentic coding environment to write and revise the simulation code, to run the studies, and to produce the tables and figures from the cached results. It was also used to draft and revise manuscript text against an outline and instructions specified by the author, and to check the drafted text against the cached results.

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Online Supplement

Supplementary tables

Draws per tail

Table S1. Expected number of simulated values per tail of the cutoff interval

Width	B = 100	B = 250	B = 400	B = 1000	B = 2000	B = 5000
0.900	5	12.5	20	50	100	250
0.950	2.5	6.25	10	25	50	125
0.990	0.5	1.25	2	5	10	25
0.995	0.25	0.625	1	2.5	5	12.5
0.999	0.05	0.125	0.2	0.5	1	2.5

Note. Computed as $(1 - w)B/2$, the average over the two tails. A highest density interval of a skewed distribution splits the excluded values unevenly, so one tail holds somewhat more and the other somewhat less. Values below roughly 5 mean that the interval endpoints are poorly estimated and the realized interval is narrower than its nominal width.

Interval convergence by condition

Table S2. Median cutoff interval width, relative to 10 000 iterations

Width	n	B = 100	B = 200	B = 400	B = 1000	B = 2000	B = 5000	B = 10 000
0.900	150	0.966	0.978	0.984	0.994	0.997	0.997	1
	250	0.968	0.980	0.988	0.996	0.996	1.000	1
	500	0.966	0.977	0.983	0.989	0.994	0.994	1
	1000	0.959	0.976	0.984	0.992	0.992	1.000	1
0.950	150	0.965	0.976	0.987	0.992	0.997	1.000	1
	250	0.959	0.976	0.983	0.990	0.995	0.997	1
	500	0.961	0.976	0.986	0.990	0.995	1.000	1
	1000	0.966	0.979	0.986	0.993	1.000	1.000	1
0.990	150	0.925	0.957	0.980	0.988	0.994	0.998	1
	250	0.925	0.957	0.981	0.990	0.995	0.997	1
	500	0.919	0.954	0.978	0.985	0.993	0.996	1
	1000	0.924	0.956	0.979	0.990	0.995	1.000	1
0.995	150	0.894	0.940	0.966	0.987	0.995	0.998	1
	250	0.892	0.938	0.966	0.986	0.993	0.998	1
	500	0.889	0.938	0.965	0.986	0.990	0.997	1
	1000	0.890	0.938	0.964	0.986	0.995	1.000	1
0.999	150	0.763	0.838	0.906	0.963	0.981	0.995	1
	250	0.762	0.838	0.907	0.961	0.980	0.995	1
	500	0.762	0.841	0.907	0.962	0.983	0.997	1
	1000	0.762	0.840	0.910	0.961	0.982	0.996	1

Note. By interval width, number of iterations, and sample size. Values below 1 indicate an interval narrower than its nominal width. Detail behind Figure 1.

Familywise false positive rates by condition

Table S3. Familywise false positive rate under the null conditions

Item set	Items	Width	Pred.	100	200	400	1000	2000	5000	10000
9 phq9	9	0.900	61.3	63.1	NA	57.8	NA	56.4	NA	NA
		0.950	37.0	43.9	NA	35.3	NA	30.3	NA	NA
		0.990	8.6	16.9	NA	10.0	NA	7.5	NA	NA
		0.995	4.4	15.0	NA	8.9	NA	5.3	NA	NA
		0.999	0.9	15.0	NA	3.9	NA	0.8	NA	NA
15 polytomous	15	0.900	79.4	85.8	NA	77.5	NA	76.2	NA	NA
		0.950	53.7	59.2	NA	55.8	NA	52.1	NA	NA
		0.990	14.0	27.1	NA	18.3	NA	13.3	NA	NA
		0.995	7.2	21.2	NA	10.8	NA	8.3	NA	NA
		0.999	1.5	21.2	NA	5.4	NA	1.7	NA	NA
20 dichotomous	20	0.900	87.8	91.0	89.8	88.9	87.9	87.0	86.1	85.9
		0.950	64.2	71.9	68.2	65.8	63.6	62.9	61.6	61.1
		0.990	18.2	37.0	28.1	22.4	18.6	17.8	16.4	16.0
		0.995	9.5	30.1	19.2	14.5	9.7	8.8	8.2	7.6
		0.999	2.0	30.1	15.7	7.7	3.4	2.6	2.0	1.6

Note. Percentages by item set, interval width, and number of iterations, with the prediction of Equation 1 (Pred.). Column headings give B. The polytomous sets were run at B = 100, 400 and 2000, and the dichotomous set at all seven values. Detail behind Figure 2.

Decision rules by condition

Table S4. Decision rules at the anchored misfit level

Rule	Items	n	Det 100	Det 400	Det 2000	FP 100	FP 400	FP 2000
Interval .999	phq9	150	28.3	21.1	14.7	2.5	0.6	0.1
		250	44.2	31.9	25.6	3.6	1.1	0.4
		1000	90.3	85.3	85.0	8.2	3.1	1.5
	sim15	250	45.0	32.3	29.4	2.8	0.8	0.3
		1000	85.2	76.9	74.0	5.1	2.0	0.7
Marginal p	phq9	150	47.5	48.9	48.9	6.4	5.7	5.7
		250	63.1	62.2	62.8	6.7	6.6	6.7
		1000	97.8	98.1	98.1	16.4	16.5	16.5
	sim15	250	62.3	62.1	61.7	5.6	5.7	5.6
		1000	92.5	92.5	93.1	10.9	11.1	10.5
Holm	phq9	150	0.0	25.0	28.1	0.0	0.3	0.3
		250	0.0	40.0	40.6	0.0	0.7	0.8
		1000	0.0	88.6	91.1	0.0	2.6	3.4
	sim15	250	0.0	31.0	37.1	0.0	0.2	0.2
		1000	0.0	74.8	81.9	0.0	0.6	1.0
WY	phq9	150	27.2	27.5	29.2	0.4	0.3	0.2
		250	41.1	43.9	41.9	1.4	0.8	0.9
		1000	91.9	91.1	91.1	4.4	3.9	3.9
	sim15	250	38.8	37.3	37.1	0.4	0.3	0.3
		1000	82.5	82.9	82.9	1.2	1.2	1.1
BH	phq9	150	1.7	25.6	28.6	0.2	0.5	0.4
		250	9.2	42.2	42.8	1.2	1.4	1.8
		1000	61.4	93.3	93.9	4.9	6.3	6.9
	sim15	250	2.3	34.4	39.0	0.3	0.4	0.6
		1000	39.0	85.4	85.4	2.0	2.6	2.9

Note. By item set, sample size, and number of bootstrap iterations. Column headings give the metric and the value of B. Det is the detection rate over the misfitting items and FP the false positive rate over the fitting items, both in percent and averaged over the misfitting-item conditions within each cell. WY is the Westfall-Young step-down and BH the Benjamini-Hochberg procedure. Holm is computed from the marginal p-values of each analysis and is not shown in Figure 3. Detail behind Figure 3.

Error rates when some items misfit

Table S5. Familywise error over fitting items and false discovery rate, conditions with misfitting items

Items	k	n	FWER Holm	FWER WY	FWER BH	FDR Holm	FDR WY	FDR BH
9 phq9	1	150	3.3	2.5	4.2	2.5	1.7	2.9
	1	250	6.7	8.3	11.7	4.2	6.2	7.0
	1	1000	21.7	25.0	34.2	11.0	12.6	18.3
	2	150	0.8	0.8	2.5	0.8	0.8	1.7
	2	250	4.2	4.2	10.0	2.8	2.4	5.1
	2	1000	26.7	30.0	50.0	10.3	11.8	19.5
15 polytomous	1	250	2.5	5.0	6.7	1.3	3.3	3.3
	1	1000	4.2	4.2	11.7	2.1	2.1	6.0
	3	250	3.3	2.5	8.3	1.4	1.1	3.4
	3	1000	20.8	22.5	42.5	6.2	7.0	15.1
20 dichotomous	1	150	2.4	2.4	5.6	1.2	1.2	2.9
	1	250	3.2	4.0	10.4	1.6	2.0	5.3
	1	500	9.2	11.2	18.4	4.6	5.6	9.5
	1	1000	26.0	28.0	44.0	13.6	14.7	25.2
	3	150	3.2	3.6	20.0	0.8	1.0	6.2
	3	250	12.4	14.4	40.4	3.3	3.9	12.9
	3	500	47.2	56.8	88.4	14.7	17.2	39.7
	3	1000	94.8	96.8	100.0	42.9	45.7	70.0
9 phq9, random responding	1	250	NA	2.5	4.2	NA	1.2	2.1
	1	1000	NA	13.3	26.7	NA	6.9	14.1
	2	250	NA	15.8	21.7	NA	6.7	9.7
	2	1000	NA	82.5	95.8	NA	38.1	52.8

Note. Percentages at $B = 2000$. FWER is the percentage of replications in which at least one fitting item was flagged, and FDR the mean proportion of flagged items that were fitting items, counted as 0 when no item was flagged. k is the number of misfitting items. The polytomous sets are at the anchored misfit level ($\rho = .85$, 120 replications), the dichotomous set at $\rho = .15$ (250 replications), and random responding at matched infit (120 replications). Holm is not available for the random-responding conditions, whose marginal p-values were not stored. WY is the Westfall-Young step-down and BH the Benjamini-Hochberg procedure.

Infit and outfit by condition

Table S6 gives the design of the sixth study together with the misfit it injects on both scales and the false positive rates it produces. The injected infit is comparable across the two item sets by construction, which is what makes the detection comparison in Table 4 interpretable. The injected outfit is not comparable, being substantially larger in the dichotomous set, and larger for the off-

target item than the on-target one within each set. Detection nonetheless runs the other way, which is the point made in the section on conditional outfit.

Table S6. Design and false positives of the infit versus outfit study

Items	Condition	n	Location	Infit	Outfit	FP infit	FP outfit	FP union
20 dich	Null	250	-	-	-	0.18	0.20	0.38
		600	-	-	-	0.24	0.12	0.36
	On target	250	0.04	1.31	1.61	0.15	0.15	0.29
		600	0.04	1.30	1.58	0.19	0.06	0.25
	Off target	250	1.82	1.25	1.84	0.13	0.13	0.25
		600	1.82	1.24	1.78	0.32	0.08	0.40
9 phq9	Null	250	-	-	-	0.53	0.40	0.76
		600	-	-	-	0.71	0.53	0.98
	On target	250	−0.15	1.27	1.32	0.50	0.10	0.55
		600	−0.15	1.28	1.32	1.30	0.75	1.45
	Off target	250	1.40	1.26	1.38	0.70	0.25	0.80
		600	1.40	1.27	1.40	1.30	0.75	1.65

Note. Location is the item location in logits relative to the person mean. Infit and Outfit are the mean observed values of the misfitting item. FP is the percentage of fitting items flagged, and Union flags an item when either statistic is significant. The null rows have no misfitting item, so their FP columns are the per-item false positive rates under a true model. Detail behind Table 4.

Monte Carlo error of the reported estimates

The replication counts differ between studies because the studies estimate different quantities with different precision requirements. Table S7 reports the Monte Carlo standard error of the key estimate from each study, computed from the cached simulation results. Proportions over independent replications use the binomial standard error. The interval convergence medians and the stability percentages use a cluster bootstrap that resamples replications or datasets, which respects the dependence of items within a dataset. The rule comparisons in the PHQ-9-anchored misfit study are paired within replications, since all decision rules are computed from the same bootstrap, and their standard errors are correspondingly small. Two consequences for interpretation follow. First, every directional claim in the paper rests on a contrast that is at least three times its standard error, or on a pattern replicated across several conditions. Second, the per-condition stability percentages are the least precise estimates in the paper, which is why the Decision stability section pools them across item sets and sample sizes and the per-condition values in Table S8 are descriptive.

Table S7. Monte Carlo standard errors of key estimates, by study

Study	Replications	Quantity	Estimate	SE
Interval convergence	500	Relative interval width	0.980	0.001
Error rates over (w, B)	500 / 250	Familywise error rate	.05 (nominal)	0.010
PHQ-9-anchored misfit	120	Detection gain, FWER vs interval	0.0 to 18.3	0.0 to 3.5
Decision stability (pooled)	10 x 25 seeds	Unstable items (%)	1.0 to 3.3	0.7
Decision stability (per condition)	10 x 25 seeds	Unstable items (%)	0.7 to 12.2	0.6 to 3.0
Decision stability (disagreement)	10 x 25 seeds	Two-analyst disagreement (%)	4.0 to 22.6	1.9 to 3.4
Mechanism check	120	False positive contrast	17.7	1.7
Infit versus outfit	250	Detection gap, infit vs outfit	4.0 to 40.8	1.2 to 3.5
Short scale	250	False positive gap, BH vs WY	2.4 to 6.4	0.6 to 0.8
Calibration	60	Mean infit of misfitting item	1.32	0.02

Note. Convergence: cluster bootstrap over replications, worst case shown (w = .999, B = 2000, n = 250). Error rates: binomial SE at the stated proportion with 500 replications, the null replications per sample size. PHQ-9-anchored misfit: paired per-replication SE of the detection gain of the FWER p-value over the interval rule, range over the ten conditions at the anchored misfit level. Stability: dataset-cluster bootstrap of the pooled and per-condition unstable-item percentages at the anchored misfit level. Disagreement: the same bootstrap applied to the probability that two seeds disagree about at least one item (Equation 5), range over the four iteration counts. That marginal standard error overstates the uncertainty of the comparison between iteration counts, which is paired within dataset: the decline from B = 400 to B = 2000 is 5.9 points with a standard error of 1.6. Mechanism: SE of the false positive contrast between misfit mechanisms. Infit versus outfit: paired per-replication SE of the detection gap between the two statistics, range over the eight conditions. Short scale: paired per-replication SE of the false positive gap between Benjamini-Hochberg and Westfall-Young at five items, range over the two sample sizes. Calibration: SE of the mean infit at the calibrated level.

Decision stability by condition

Table S8. Seed-dependent flags by condition

Rule	Item set	Correlation	n	B = 100	B = 400	B = 1000	B = 2000
FWER p-value	phq9	0.15	250	30.0	20.0	16.7	12.2
			1000	0.0	0.0	0.0	0.0
		0.50	250	23.3	12.2	11.1	8.9
			1000	16.7	12.2	11.1	6.7
		0.85	250	6.7	3.3	2.2	1.1
			1000	12.2	6.7	3.3	2.2
		none (null)	250	2.2	0.0	0.0	0.0
			1000	1.1	0.0	0.0	0.0
	sim15	0.15	250	40.0	16.0	8.7	8.0
			1000	0.0	0.0	0.0	0.0
		0.50	250	12.7	3.3	2.0	1.3
			1000	32.0	17.3	9.3	6.7
		0.85	250	4.7	1.3	1.3	0.7
			1000	7.3	3.3	2.0	0.7
		none (null)	250	1.3	0.7	0.7	0.0
			1000	0.7	0.0	0.0	0.0
Interval, w = .999	phq9	0.15	250	37.8	28.9	20.0	20.0
			1000	1.1	1.1	1.1	2.2
		0.50	250	42.2	30.0	20.0	14.4
			1000	40.0	37.8	27.8	28.9
		0.85	250	13.3	7.8	5.6	4.4
			1000	26.7	18.9	12.2	12.2
		none (null)	250	5.6	2.2	2.2	2.2
			1000	5.6	0.0	0.0	0.0
	sim15	0.15	250	48.0	36.0	25.3	22.7
			1000	6.0	8.7	8.0	6.7
		0.50	250	32.7	20.7	12.7	10.7
			1000	44.7	46.0	31.3	28.0
		0.85	250	18.7	6.0	4.0	2.7
			1000	22.0	12.0	7.3	6.7
		none (null)	250	4.7	2.0	0.7	0.7
			1000	5.3	1.3	0.0	0.7

Note. Percentage of items with a seed-dependent flag across 25 bootstrap seeds (10 datasets per condition). Values carry Monte Carlo standard errors of roughly 1 to 3 percentage points, see the section above.

Reproducibility

Everything needed to reproduce this paper is in the repository at https://github.com/pgmj/rasch_fwer, which contains the manuscript source, the simulation code, and the cached simulation results.

The manuscript is written as a Quarto document (Allaire et al., 2026). Every number reported in the body, including those in the figures and tables, is computed at render time from the cached results rather than transcribed, so the reported values cannot drift from the data behind them. The abstract is the one exception: Quarto evaluates the document metadata before the code that loads the caches, so the few figures quoted there are typed and were checked against the body by hand. Rendering the manuscript reproduces the whole paper in seconds without rerunning any simulation. This is also relevant to assessing the use of large language models described in the main text. No reported number depends on a model or a person correctly copying it, and any reported quantity can be recomputed independently of the process that produced it.

The studies are in `simulations/`, each writing its results to an `.rds` file beside it, apart from the two dichotomous studies, which were rerun for the revision with fresh data in every replication and are in `rev1/` (`fresh_rerun.R`). The original master-dataset versions of those two studies remain in `simulations/` for comparison. Three further scripts hold the calibration scans that fix the misfit levels, two of which the manuscript reads directly for Table 2 and for the five-item calibration in the Method. Shared code, including the data generators and the plot theme, is in `R/helpers.R`. The seven studies comprise 18 180 design cells and took about 67 hours to run on ten M4 cores. Each study uses a resumable runner. Design cells are keyed on the columns of the design grid, results are checkpointed to disk after every chunk, and cells already present in the cache are skipped, so raising a replication count and rerunning computes only the new cells. Seeds are deterministic functions of the design cell rather than of the execution order, which means any single cell can be reproduced on its own without running the study that contains it.

All results were produced with R 4.6.1 (R Core Team, 2026) and `easyRasch2` (Johansson, 2026a), under versions 1.1.0 and 1.1.1 and, for the two dichotomous studies, 1.4.0. As a check, a cell of the original dichotomous study rebuilt under 1.4.0 reproduced its p-values and interval flags to the last digit. The package computes conditional item fit through `iarm` 0.4.3 (Mueller & Santiago, 2022) and estimates item parameters by conditional maximum likelihood with `psychotools` 0.7.6 (Zeileis et al., 2026). The multidimensional dichotomous data were generated with `eRm` 1.0.10 (Mair & Hatzinger, 2007), the polytomous data with the generators in `easyRasch2`, and the highest density intervals computed with `ggdist` 3.3.3 (Kay, 2025). The phq9 data used to calibrate misfit severity are included in `easyRasch2`.